DAIM SHORT COURSE · INTRODUCTION TO MACHINE LEARNING · WEEK 2

Applying Logistic Regression

One real decision, worked from start to finish with a classifier that gives a probability

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How to use this reading

This is the pre-reading for the second live session. It tells the whole Week 2 story on its own: you do not need the slides, any code, or any maths beyond reading a chart and adding up a few numbers. Set aside about forty-five minutes.

One example runs through the whole document: a **screening clinic** deciding which people to refer for a confirmatory test for **type 2 diabetes**. The problem is real; every patient in it is invented. We will ask why it is a machine-learning problem, meet a specific model called **logistic regression**, watch it being fitted, and then apply it by hand to individual patients.

Shaded boxes like this one are worth pausing on. The questions at the end are the ones we will discuss in the webinar, so try them before you arrive.

The story in one paragraph. Last week we saw that machine learning replaces a hand-written rule with a rule learned from labelled examples, and that a classifier draws a boundary between two groups. This week we take one real decision and build that classifier properly. We start with a clinic that must decide who to send for a blood test, and ask why a learned rule is better than a written one. We then open up logistic regression, which does three simple things: it adds up the clues with weights, squashes the total into a probability between 0 and 1, and compares that probability with a threshold. We watch the weights being fitted, read what the fitted model has learned, and apply it to six people one at a time. Two of the six are mistakes of different kinds, and that is where next week begins.

1. Where we got to

Four ideas from Week 1 carry the whole of this week, so here they are in one breath each.

- **Tasks.** Classification predicts a category; regression predicts a number. Today's task is classification.
- **Experience.** Supervised learning needs labelled examples: rows with the answer already known. Some of those rows are kept aside as a test set that the model never sees while it learns.
- **Learning means fitting.** A model has adjustable dials, its parameters. Fitting turns the dials until the model's answers match the known answers as well as they can.
- **The boundary.** For "will the customer buy?", the fitted model divided the age–salary picture into two regions with a line between them. Last week we also noticed that a model which gives a *score*, not just a side of the line, carries more information. This week's model gives exactly that score.

Everything that follows builds on these four. Same picture, different problem.

2. A screening clinic

2.1 The problem

Diabetes UK reports that more than 4.7 million people in the UK had been diagnosed with diabetes in 2024–25, around nine in ten of them with type 2, and estimates that almost 1.3 million more have type 2 diabetes without knowing it [1]. That second group is the reason screening exists. Type 2 diabetes develops quietly, it is far easier to manage when caught early, and the test that confirms it is a routine blood test. The difficulty is deciding who should have that test.

Picture a screening clinic. It sees people who have already been flagged as at risk, perhaps by a GP or a community programme, and at each visit it records a handful of things that are easy to measure: the person's **age**, their **BMI**, a **fasting glucose** reading from a finger-prick test, and whether they have a **family history** of the condition. On the basis of those four clues the clinic must decide, for each person: refer for a confirmatory blood test, or not. Referring everyone would be expensive and would alarm a great many healthy people; referring no one would leave the undiagnosed undiagnosed.

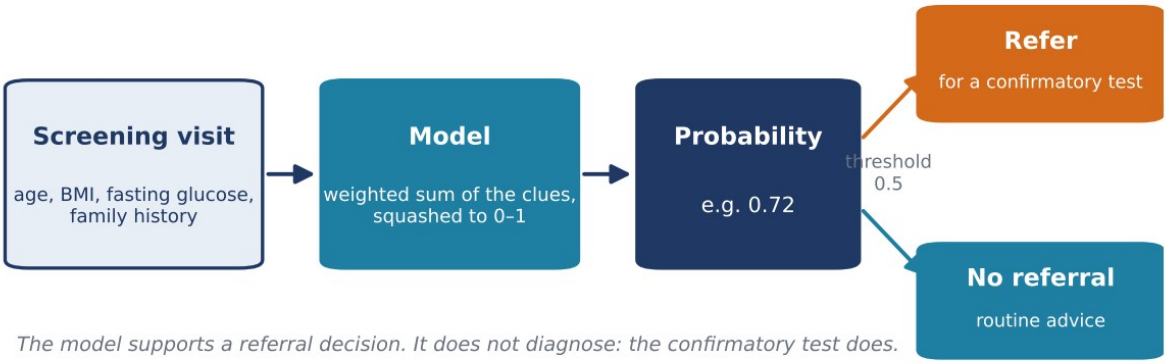


Figure 1. The screening decision. Four clues are recorded at the visit; a model turns them into a probability; a threshold turns the probability into a decision. Each step on this picture gets its own section below.

One sentence to keep in mind for the rest of the course: **the model supports a referral decision; it does not diagnose.** Diagnosis is what the confirmatory test does. The model's job is narrower and more modest: to help the clinic spend its tests where they are most likely to matter.

2.2 Why machine learning? Why not just write a rule?

Week 1 gave us a test to apply to any proposed machine-learning project: is there a clean rule a person could write down, and is there labelled experience to learn from instead? Apply it here.

A natural first rule is a single glucose cut-off: refer anyone whose fasting glucose is above some number. In real practice a laboratory fasting glucose of 7.0 mmol/L or more is in fact one of the formal diagnostic criteria [2], so the idea is not silly. But the screening visit records one finger-prick reading, which is noisier than a laboratory result, and glucose is only one of the four clues. Some people with the condition have a screening reading below any cut-off you choose, and some healthy people have a reading above it. Age, BMI and family history all shift the picture, and they shift it *together*: a reading of 6.4 means something different in a 74-year-old with a family history than in a 46-year-old without one. Nobody can sit down and write out the right combination by hand.

What the clinic does have is experience. Past patients were screened, then had the confirmatory test, and the result was recorded. For each of them we know the four clues *and* the true answer. That is precisely the labelled experience supervised learning needs, and the pattern in it is learnable: people with the condition tend to have higher glucose and higher BMI, to be older and to have a family history, but with a great deal of overlap. A model can learn how much weight each clue deserves, which is exactly the thing we could not write down ourselves.

2.3 Why classification?

The data look like this. Here are five of the 1,000 invented patients in the course dataset.

Patient	Age	BMI	Fasting glucose (mmol/L)	Family history	Has type 2 diabetes?
P0003	35	31.5	6.0	no	no
P0004	71	32.8	7.0	yes	yes
P0006	53	31.6	4.3	no	no
P0007	53	29.0	4.5	no	no
P0008	59	29.1	5.8	yes	no

The first four columns are the features: what the clinic knows at the visit. The last column is the label: what the confirmatory test later showed. Invented patients, generated for this course.

The label takes one of two values, so this is **classification**, just as spam-or-not-spam was. (Had the clinic wanted to predict a person's laboratory glucose *value*, that would have been regression; same clues, different target.) Three numbers describe the dataset. There are **1,000** patients, each with **4** clues, and about **1 in 10** has the condition. That last figure is higher than in the general population because the clinic only sees people already flagged as at risk, and it will matter a great deal next week.

Not all 1,000 are used for learning. Following the rule from Week 1, **600** of them form the training set that the model learns from, and the other **400** are locked away before fitting begins and never shown to the model while it learns. They come back next week, when we ask how well the model does on people it has never seen. Everything in Sections 3 and 4 uses the 600 alone, apart from the six individual patients in Section 4.4, who are drawn from the 400.

2.4 Looking at the data

Before fitting anything, look at the 600 training patients. The left panel of Figure 2 uses only glucose; the right panel uses glucose and BMI together.

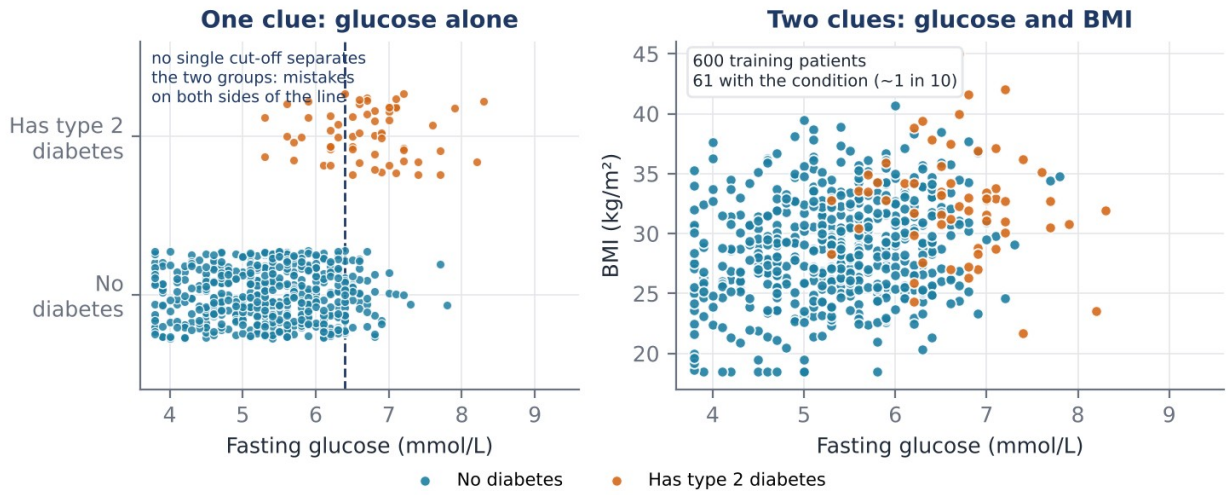


Figure 2. The 600 training patients. Left: fasting glucose alone, with each patient placed in the row for their true outcome. Right: glucose and BMI together. Orange marks patients who turned out to have the condition, teal those who did not. Invented data.

The left panel is the single-cut-off rule made visible. Wherever you place the vertical line, there are orange dots to its left (people with the condition who would be sent home) and teal dots to its right (healthy people who would be referred). The right panel is more hopeful: the orange patients gather towards the upper right, where glucose and BMI are both high, so the two clues together separate the groups better than either alone. But "better" is not "perfectly". The groups overlap, and no straight line, or any other line, will separate them cleanly. That overlap is not a flaw in the invented data; it is what real screening data look like. A good model will still make mistakes, and the honest question is how many and which kind.

> **Which patients are drawn, and which are not**
>
> The pictures in this reading use two of the four clues, glucose and BMI, because a two-feature picture can be drawn on a page. The model we actually fit uses all four. Everything you see in the two-feature pictures, the line, the shaded regions, the probabilities, happens in exactly the same way with four features; it simply happens in a space we cannot draw. Keep that in mind whenever a figure looks simpler than the table beside it.

3. Logistic regression: a classifier that gives a probability

3.1 The idea in one sentence

Here is the whole model: **add up the clues, each with its own weight; squash the total into a number between 0 and 1; compare it with a threshold.** Three moves, and each has a name.

1. **Weighted sum (the score).** Each clue is multiplied by a weight and the results are added to a starting point. The total is called the score.
2. **Squash.** The score can be any number, large or small, positive or negative. A fixed S-shaped curve converts it into a probability between 0 and 1.

3. **Threshold.** If the probability is above a chosen cut-off, 0.5 to begin with, predict "has the condition"; otherwise predict "does not".

Written out, the first move is a single line. With four clues x_1, x_2, x_3, x_4, four weights w_1, w_2, w_3, w_4 and a starting point b, the score z is

$$z = b + w_1x_1 + w_2x_2 + w_3x_3 + w_4x_4$$

where each x is a clue (a patient's glucose, say), each w is that clue's weight, and b is the score a patient would have if every clue contributed nothing. The weights and the starting point are the **dials** from Week 1: they are what fitting adjusts, and they are the entire model. The name is a historical accident worth decoding once. The weighted sum inside the model is the same kind of straight-line formula we fitted to salaries last week, which is regression; and the probability it produces is a number. But the model's *output* is a yes-or-no decision, which makes it a classifier. So: regression in the name, classifier in behaviour.

3.2 The weighted sum: what the dials mean

Think of the score as a running total that starts at b and moves every time a clue is read. A clue with a large positive weight pushes the total up a lot per unit; a clue with a small weight barely moves it; a clue with a negative weight would pull it down. Two things follow immediately.

First, a weight is always *per unit of its clue*, so weights can only be compared once the units are taken into account. A weight of 0.18 per BMI unit and a weight of 0.99 for family history are not saying that family history matters five times more than BMI; five BMI units add 0.9, about the same as a family history. Second, the starting point b on its own is not very meaningful, because no real patient has an age of zero or a BMI of zero. The useful reference is a *typical* screened patient. In Section 4.2 we will start from such a patient and read every other patient as a departure from them.

For the moment the key fact is that the score is only a running total. A score of +7 or −12 means nothing to a clinician on its own. Which is why the second move exists.

3.3 The squash: the logistic function

The S-shaped curve in Figure 3 is called the **logistic function**, and it is where the model gets its name. Feed it any score and it returns a number strictly between 0 and 1. A score of 0 returns exactly 0.5. Large positive scores return numbers close to 1; large negative scores return numbers close to 0; scores of +2 and −2 return 0.88 and 0.12.

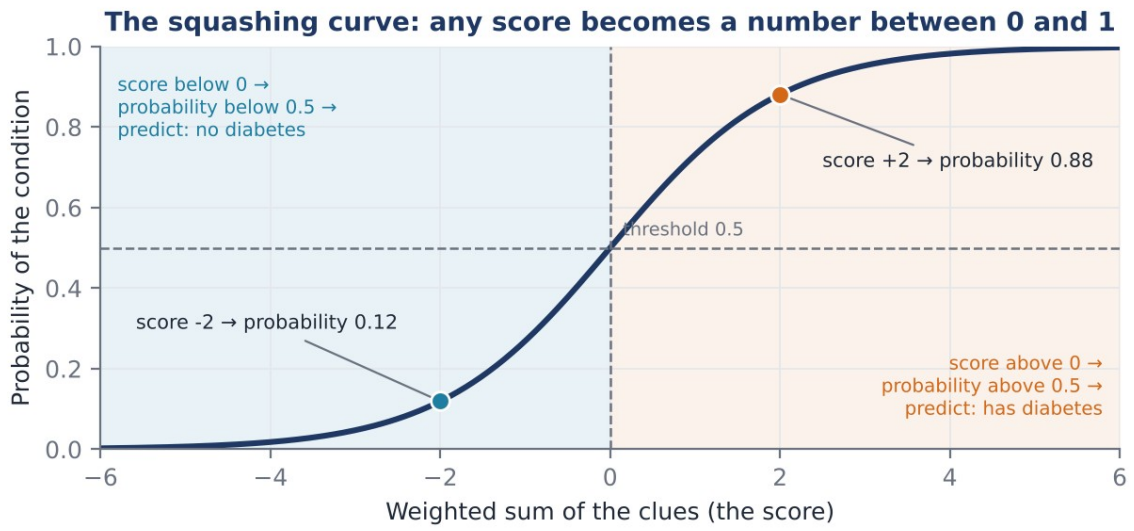


Figure 3. The squashing curve. Any score on the horizontal axis becomes a probability on the vertical axis. A score of 0 lands exactly on 0.5; +2 gives 0.88 and −2 gives 0.12.

Why bother? Because a probability is something people can act on. "0.9" and "0.02" mean something to the clinic in a way that "+7" and "−12" never will. And notice something important about the curve: **it is fixed**. It is not learned, it never changes, and it is the same for every logistic regression model ever fitted. The only things the data get to change are the weights. That is what keeps the model simple enough to read.

3.4 The threshold and the decision boundary

The third move turns the probability into a decision. Choose a threshold, 0.5 to begin with, and predict "has the condition" for anyone at or above it. Because the curve passes through 0.5 at a score of exactly 0, a threshold of 0.5 on the probability is the same as a threshold of 0 on the score: positive score, refer; negative score, do not.

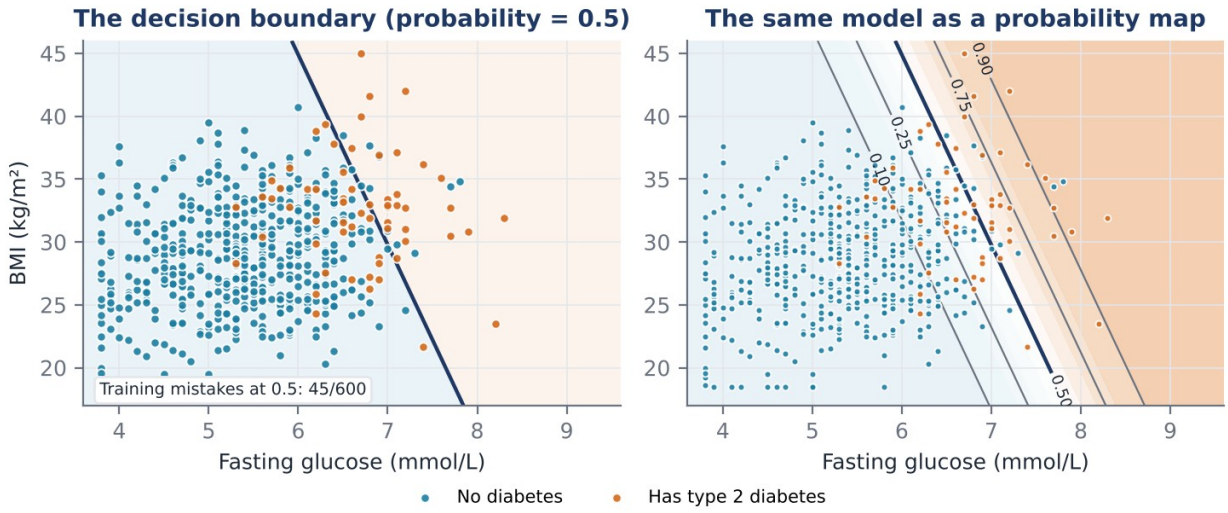


Figure 4. Left: the two-clue model with a threshold of 0.5. The navy line is the decision boundary; the orange tint is where the model predicts the condition. 45 of the 600 training patients land on the wrong side. Right: the same model drawn as a probability map, with the 0.10, 0.25, 0.50, 0.75 and 0.90 contours. The boundary is just the 0.5 contour.

In the two-clue picture the boundary comes out as a straight line, because the score is a weighted sum and a weighted sum of two clues equals zero along a line. The left panel shows where that line

fell after fitting: 45 of the 600 training patients are on the wrong side of it, most of them orange dots in the teal region. The right panel is the same model drawn more honestly. Every point in the picture has a probability, not merely a side. Patients just to the left of the line are near 0.45; patients far to the left are near 0.01. The line is simply the place where the probability crosses 0.5, and if we moved the threshold the line would move with it.

3.5 Why a probability is more useful than a verdict

It would be possible to build a classifier that only says yes or no. Logistic regression gives a number as well, and the number is worth having for three reasons.

- **It ranks people.** Two patients both predicted "no" are not the same if one is at 0.02 and the other at 0.45. The clinic can see who is close to the line, and might choose to look at those people again.
- **It is honest about doubt.** A probability of 0.55 says "barely". A verdict of "yes" hides that. Communicating uncertainty is part of using a model responsibly, and a model that cannot express it forces people to pretend to a confidence nobody has.
- **The threshold is a choice.** 0.5 is only a starting point. Lower it and more people are referred; raise it and fewer are. The *model* has not changed; the *decisions* have. Next week is largely about making that choice well.

> **Pause and think**
>
> Before reading on: the clinic's two possible mistakes are referring a healthy person for a test they did not need, and sending home a person who has the condition. Which of these would you rather the model made more often? Does your answer suggest a threshold above 0.5, or below it?

4. Fitting, then applying

4.1 Turning the dials until the loss stops falling

We now know what the dials are. How are they set? The same way the salary line was set last week: by measuring how wrong the model is, nudging the dials in the direction that makes it less wrong, and repeating. The measure of wrongness is the **loss**. For a classifier that outputs probabilities the natural loss is this: for each training patient, how far was the probability the model gave from the truth? Giving 0.9 to a patient who has the condition costs almost nothing; giving 0.1 to that same patient costs a great deal. A confident wrong answer is punished much more than a hesitant one. Add up those costs over all 600 training patients and you have the loss.

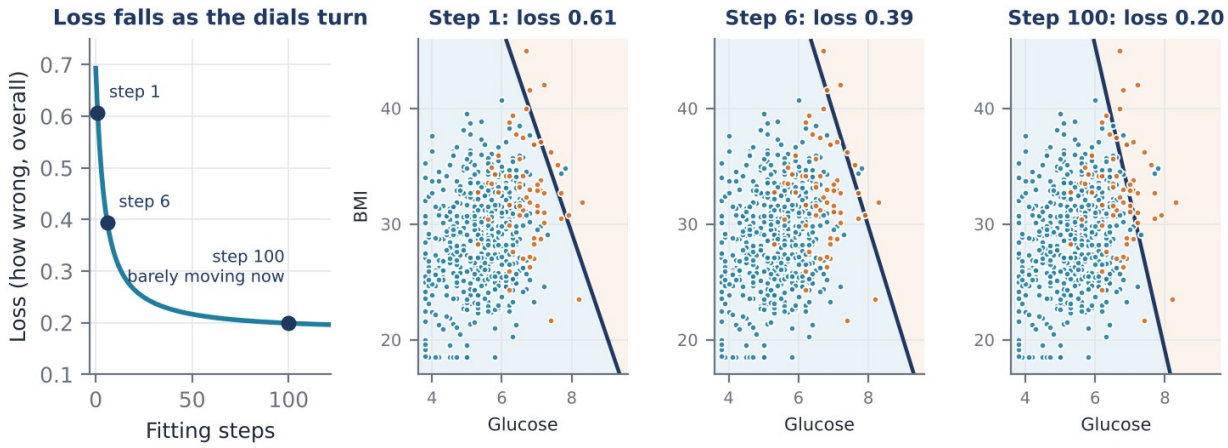


Figure 5. Fitting the two-clue model. Left: the loss falls with each adjustment of the dials. Right: where the decision boundary sat after 1, 6 and 100 steps. Within a hundred steps it has essentially settled.

Figure 5 shows the process for the two-clue model. The dials begin at zero, which gives every patient a probability of exactly 0.5; the loss starts at 0.69. One step later it is 0.61 and the boundary has already swung towards the orange corner. After six steps the loss is 0.39; after a hundred it is 0.20, and from there it barely moves. No calculus has been shown here, and none is needed to understand what is happening: each step is a small turn of every dial in the direction that lowers the loss, and the fitting stops when no small turn helps any more. Compare this with Week 1's salary line, which could be calculated in one go. For logistic regression there is no such shortcut, so the dials are found by this stepwise search; but the idea, find the parameters with the lowest loss, is identical.

> **Loss and performance are different things**
>
> The loss is the quantity the *fitting* procedure minimises. It is an internal measure, chosen because it is smooth and rewards good probabilities. It is not the number we will report to the clinic. How well the finished model actually performs on patients it has never seen, and which of its mistakes matter, is a separate question with separate measures, and it is next week's subject. Fitting on the 600 and then judging on the 600 would be marking your own homework; that is why the 400 are locked away.

4.2 What the data taught the model

Here is the fitted four-clue model, the one the clinic would actually use. The weights are given both per unit, as the model stores them, and in everyday units, which is how they are easiest to read aloud.

Clue	Weight (adds to the score)	In words
Fasting glucose (w_1)	+2.52 per mmol/L	The dominant clue: one extra mmol/L raises the score by 2.5.
Family history (w_2)	+0.99 if yes	Raises the score by about 1, the same as five BMI units or twenty years of age.
BMI (w_3)	+0.18 per unit (+0.91 per 5 units)	Moderate.
Age (w_4)	+0.05 per year (+0.50 per 10 years)	Weak on its own.
Starting point (b)	−4.4 for a typical patient	Age 56, BMI 29.5, glucose 5.5, no family history, which gives a probability of about 0.01.

Fitted on the 600 training patients. The starting point is shown for a typical screened patient rather than as the raw b (which is −26.5, an uninterpretable number because it corresponds to a patient of age zero and BMI zero).

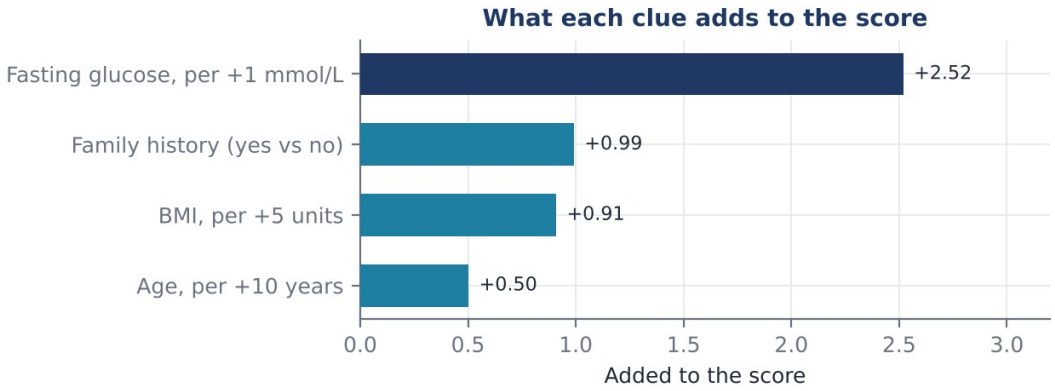


Figure 6. The same weights as bars, each in an everyday unit. Glucose dominates; family history, five BMI units and ten years of age are worth roughly 1, 0.9 and 0.5 respectively.

Nobody told the model that glucose matters most. It found that out from the 600 patients, and it found the relative sizes of the other three clues at the same time. That is the whole point of letting the data set the dials: the combination that no one could write down by hand has been written down by the fitting procedure, and it fits on one line. Four weights and a starting point are the entire model. There is nothing else inside it.

Read the typical patient carefully, because they are the anchor for everything that follows. A 56-year-old with a BMI of 29.5, a fasting glucose of 5.5 and no family history starts at a score of about −4.4, which squashes to a probability of roughly 0.01. The typical person at this clinic, in other words, is well on the "no" side. The patients who end up referred are those whose clues add up: a few extra years, a higher BMI, and above all a higher glucose reading, each pushing the running total towards and then past zero.

4.3 One patient, worked through by hand

Take a real case: Patient E, a 66-year-old with a BMI of 37.1, a fasting glucose of 6.9 mmol/L and no family history. Figure 7 adds up the clues.

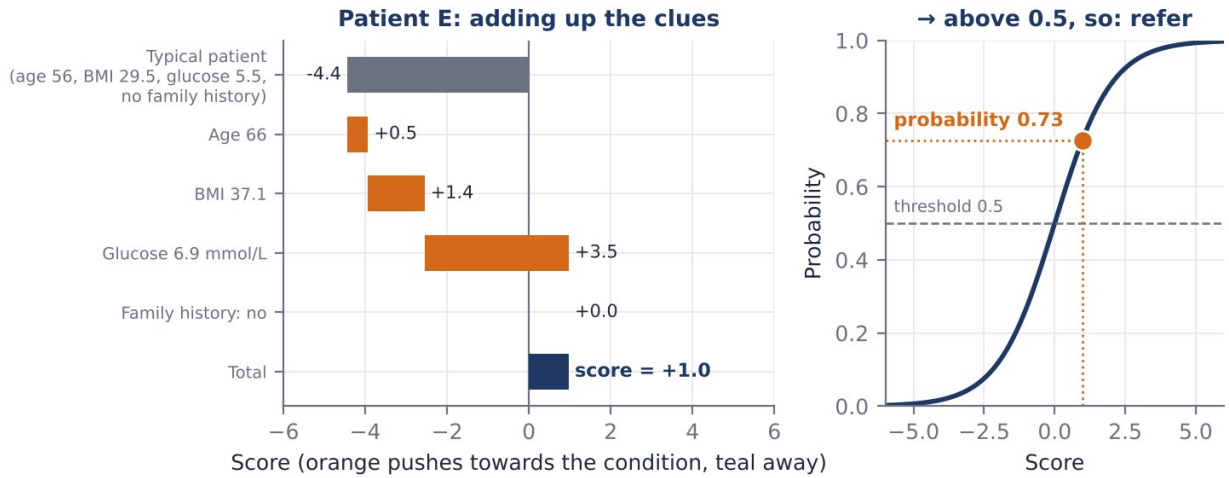


Figure 7. Patient E. Left: starting from the typical patient's score, each clue adds its contribution; orange bars push towards the condition. Right: the total of +1.0 is squashed to a probability of 0.73, which is above 0.5, so the decision is to refer.

Start at −4.4, the typical patient. Age 66 is ten years above typical, and ten years add 0.5. A BMI of 37.1 is 7.6 units above typical, and at 0.18 per unit that adds about 1.4. Glucose of 6.9 is 1.4 mmol/L above typical, and at 2.52 per mmol/L that adds about 3.5, by far the largest contribution. No family history adds nothing. The total is −4.4 + 0.5 + 1.4 + 3.5 = +1.0. Squash it: a score of +1.0 gives a probability of 0.73. That is above 0.5, so the decision is to refer. The confirmatory test later showed that Patient E does have the condition: a correct referral.

Notice what you just did. You applied a machine-learning model, by hand, to a real case, with nothing but addition and a glance at a curve. This is why logistic regression is the first classifier taught in nearly every course: not because it is the most powerful, but because every step of it can be followed and explained, to a colleague, to a patient, or to a regulator.

4.4 Six patients, side by side

Patient E is one of six patients, A to F, that we will keep returning to. None of the six was used in fitting, so their outcomes are a fair check rather than a rehearsal. The table gives their clues, the probability the fitted model assigns, the decision at a threshold of 0.5, what the confirmatory test showed, and whether the decision was right.

Patient	Age	BMI	Glucose	Family history	Probability	Decision at 0.5	Confirmatory test	Outcome
A	43	32.6	5.3	no	0.01	No referral	No diabetes	Correct
B	46	30.5	6.8	no	0.19	No referral	No diabetes	Correct
C	74	32.8	6.4	yes	0.58	Refer	No diabetes	False alarm
D	61	37.9	6.4	no	0.40	No referral	Has diabetes	Missed case
E	66	37.1	6.9	no	0.73	Refer	Has diabetes	Correct
F	58	30.0	7.7	yes	0.91	Refer	Has diabetes	Correct

Six invented patients from the 400 kept aside, chosen to span the range of probabilities. Decisions use the four-clue model with a threshold of 0.5.

Read down the table. A and B are a clear and a borderline "no", and both are right. F is a clear "yes": high glucose and a family history push the probability to 0.91. E we have just met. The two remaining rows are the ones to dwell on.

Patient C is 74, has a family history and a glucose reading of 6.4. The model gives 0.58, just over the line, and refers. The confirmatory test is negative. This is a **false alarm**: an unnecessary test, some worry, a little cost, and no lasting harm.

Patient D is 61 with a BMI of 37.9 and the same glucose reading, 6.4, but no family history. The model gives 0.40, just under the line, and does not refer. The confirmatory test is positive. This is a **missed case**: a person with the condition sent home with routine advice, and the undiagnosed group one person larger.

Look at what separates C from D. Their glucose is identical; C's family history and extra years lift their score just above zero, D's higher BMI does not quite compensate for the missing family history, and the two land 0.18 apart on either side of 0.5. The model did nothing unreasonable with either. Its probabilities, 0.58 and 0.40, both say "uncertain, near the line". The *decisions* came from the threshold, and the threshold was a choice we made without thinking about it. We said 0.5 because a score of zero was a natural place to start; we did not ask whether a false alarm and a missed case are equally bad. In a screening clinic they plainly are not. Keep Patient D in mind. Next week starts with them.

> **Pause and think**
>
> Suppose the clinic decided that missing a case is roughly four times worse than a false alarm. Would you move the threshold up or down from 0.5? What would happen to patients C and D? And what would you expect to happen to the number of referrals across all 1,000 patients?

5. Pulling it together

Week 1's three questions, asked of this week's one problem, with the pipeline from Week 1 filled in step by step.

	In general	In the screening problem
Task	Classification: predict one of two categories	Refer for a confirmatory test, or not
Experience	Past examples with known labels	Past patients: age, BMI, fasting glucose, family history, and the confirmatory test result
Features and label	Inputs available at prediction time; the answer to learn	The four clues recorded at the visit; whether the person has type 2 diabetes
Model family	A shape of rule chosen by a person	Logistic regression: weighted sum → logistic squash → threshold
Fitting	Turn the dials to minimise a loss on the training set	Four weights and a starting point, set on 600 patients until the loss stopped falling
Applying	Feed in a new case, read off the prediction	Patient E: score +1.0, probability 0.73, refer
Performance	Judged on unseen examples, with the right measure	Next week: the 400 patients kept aside, and what 0.5 is costing us

Key terms

Term	Meaning
Logistic regression	A classifier that computes a weighted sum of the features, squashes it to a probability, and compares that with a threshold.
Weight	The number a feature is multiplied by in the weighted sum; one of the model's dials. Always per unit of its feature.
Starting point (intercept, b)	The score when every feature contributes nothing; the other dial.
Score (z)	The weighted sum of the features plus the starting point. Can be any number.
Logistic function	The fixed S-shaped curve that turns a score into a probability between 0 and 1. Also called the sigmoid.
Probability	The model's output before any decision: how likely it believes the positive class to be.
Threshold	The cut-off on the probability that turns it into a decision. 0.5 by default; a separate choice from the model.
Decision boundary	The set of points where the probability equals the threshold. For logistic regression with two features, a straight line.
Loss	How wrong the model's probabilities are across the training examples; what fitting minimises. A confident wrong answer costs more than a hesitant one.
False alarm	A positive prediction for a case that is actually negative (Patient C).
Missed case	A negative prediction for a case that is actually positive (Patient D).

Check yourself before the webinar

Answer each in your own words first, then compare with the pointer underneath.

1. In the screening table, which columns are features and which is the label? Why is this classification and not regression, even though the glucose column is a number?

> Age, BMI, fasting glucose and family history are the features; "has type 2 diabetes?" is the label. It is classification because the *label* takes one of two categories. The features can be numbers; what decides the task is the kind of thing being predicted.

2. A patient's clues add up to a score of exactly 0. What probability does the model give them, and what is the decision at a threshold of 0.5? What if the threshold were 0.6?

> A score of 0 squashes to a probability of exactly 0.5, so at a threshold of 0.5 they are referred (just). At 0.6 they would not be. The model's output is the same in both cases; only the decision changed.

3. The weight for family history is 0.99 and the weight for BMI is 0.18. A colleague concludes that family history matters about five times more than BMI. What is wrong with that reading?

> Weights are per unit of their feature. 0.18 is per single BMI unit; five BMI units add 0.9, about the same as a family history. Weights can only be compared once units are taken into account, which is why the reading converts them to everyday units.

4. Patient D has the condition and the model gave 0.40. Did the model "get it wrong"? What exactly produced the missed case?

> The model gave a probability that said "uncertain, a little below the line". The missed case was produced by the threshold of 0.5, which turned 0.40 into "no referral". Whether 0.5 is the right threshold for a screening clinic, where a missed case costs more than a false alarm, is next week's question.

5. Why were 400 patients locked away before fitting, and why does it matter that none of A to F were among the 600?

> The loss on the 600 training patients tells us how well the model fits the past, not how it will do on new people. Only patients the model never saw while learning can tell us that. If A to F had been in the training set, their outcomes would be a rehearsal rather than a check.

> **Next week**
>
> We let the fitted model loose on the 200 of the 400 held-out patients that form the **test set** and count its mistakes. The headline number will look good, and we will show why it should not be trusted on its own: when only 1 in 10 people has the condition, a model that says "no" to everyone is already 90 per cent accurate. We will then name the two kinds of mistake properly, meet **precision**, **recall** and **F1**, and finally do what this week deliberately left undone: choose the **threshold** on purpose, using the other 200 held-out patients, and see what each choice costs. Same model, same dials, no refitting.

Sources and reuse

This reading was written for the DAIM short course and is intended for open sharing and reuse. The screening scenario is real in outline, but every patient, every reading and every fitted number in this document is an invented teaching illustration generated for the course; the dataset is not real patient data and must not be used for any clinical purpose. The model was fitted with standard open-source software [8] and the figures were generated from that fit. The explanations draw on the following publicly accessible sources, checked 2 October 2026.

[1] Diabetes UK, How many people in the UK have diabetes? (2024–25 figures)

[2] World Health Organization, Diabetes fact sheet (diagnostic criteria, including fasting plasma glucose ≥ 7.0 mmol/L)

[3] Google, Machine Learning Crash Course: Logistic regression: calculating a probability with the sigmoid function

[4] Google, Machine Learning Crash Course: Logistic regression: loss and regularization

[5] Google, Machine Learning Crash Course: Classification: thresholds and the confusion matrix

[6] Google, Machine Learning Glossary

[7] Google, Machine Learning Crash Course: Linear regression: gradient descent

[8] scikit-learn developers, Logistic regression (user guide)

[9] scikit-learn developers, Probability calibration

[10] Google, Machine Learning Crash Course: Overfitting and generalization: dividing the original dataset

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